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Tell me about the latest trends and most researched algorithms for fine-tuning neural networks

Fine-Tuning Neural Networks: Latest Trends, Algorithms, Advancements, and Research Gaps

Quick Reference

Key Findings Table

Trend/Algorithm	Description & Impact	Domains/Applications	Key References
Parameter-Efficient Fine-Tuning	LoRA, QLoRA, adapters update a small subset of parameters, reducing cost and memory with near full performance	NLP, Vision, Remote Sensing, Edge	[1] [2] [3]
Prompt-Based Fine-Tuning	Lightweight, context-driven adaptation; excels in few-shot/zero-shot, but less robust in complex domains	NLP, Medical Imaging	[4] [5]
Hybrid Adapter-Prompt Methods	Combine adapters and prompts for improved flexibility and domain adaptation	Biomedical, Clinical NLP	[6] [7] [8]
RL-Based Fine-Tuning (RLHF, DPO)	Uses human feedback or reward models for alignment, sample efficiency, and explainability	LLMs, Medical Report Generation	[9] [10] [11]
Hardware-Aware Fine-Tuning	Co-design with RRAM/ SRAM, quantization, drift mitigation for edge/embedded deployment	Edge AI, In-memory Computing	[12] [13] [14]
Multi-Source/Ensemble Fine-Tuning	Leverages multiple datasets, ensemble strategies for robustness and generalization	Biomedical Signal Processing	[15] [16]
Privacy-Preserving Fine-Tuning	Differential privacy, federated learning, BitLoRA for confidential data adaptation	Healthcare, Finance, On-device AI	[17] [18] [19]

Trend/Algorithm	Description & Impact	Domains/Applications	Key References
Catastrophic Forgetting Mitigation	IMSM, Bayesian PEFT, Half Fine-Tuning, learning rate scheduling to retain prior knowledge	NLP, Continual Learning	[20] [21] [22]

Direct Answer

The latest trends in fine-tuning neural networks emphasize parameter-efficient methods such as LoRA, QLoRA, and adapter-based fine-tuning, which allow updating only a small subset of model parameters rather than the entire pre-trained model. Prompt tuning and reinforcement learning-based fine-tuning (e.g., RLHF and DPO) are also gaining traction for improving efficiency and model alignment. Hybrid methods that combine adapters and prompt tuning are showing promising results, especially in specialized domains. Hardware-aware fine-tuning and algorithm-hardware co-design frameworks are emerging to address deployment challenges on resource-constrained or edge devices, balancing performance with energy efficiency and latency.

Study Scope

- Time Period: 2022-2024 (focus on the most recent literature)
- Disciplines: Machine Learning, Natural Language Processing, Computer Vision, Biomedical Engineering, Edge Computing, Hardware-Software Co-Design
- Methods: Systematic reviews, meta-analyses, comparative studies, empirical benchmarks, hardware-in-the-loop experiments

Assumptions & Limitations

- The report synthesizes findings from recent peer-reviewed literature and systematic reviews; some rapidly evolving subfields (e.g., hardware-aware fine-tuning) may have additional unpublished advances.
- Comparative performance metrics are domain- and dataset-dependent; generalizations are made where consistent trends are observed.
- Privacy-preserving and catastrophic forgetting mitigation methods are still maturing, and their effectiveness may vary with model scale and application context.

Suggested Further Research

- Continual Learning: Robust, scalable solutions for catastrophic forgetting, especially in cross-domain and low-resource settings.
- Hardware-Aware Fine-Tuning: Unified frameworks for algorithm-hardware co-design, especially for nonvolatile memory and analog in-memory computing.

- **Privacy & Security:** Integration of differential privacy with hybrid adapter-prompt methods and federated learning for sensitive domains.
- **Domain Adaptation:** Efficient, automated domain adaptation and negative transfer mitigation in multi-source fine-tuning.
- **Interpretability:** Methods to enhance transparency and explainability in fine-tuned large models.

1. Introduction

Fine-tuning neural networks has become a cornerstone of modern machine learning, enabling the adaptation of large pre-trained models (foundation models) to specialized downstream tasks with limited data and computational resources. As models scale to billions of parameters, full model fine-tuning becomes increasingly impractical, driving the development of parameter-efficient, hardware-aware, and privacy-preserving fine-tuning strategies. These advances are critical for deploying AI on edge devices, in privacy-sensitive domains, and for continual learning scenarios where models must adapt without forgetting prior knowledge [8] [23] [24].

Overview of Fine-Tuning in Neural Networks

Fine-tuning leverages the general representations learned by large models and adapts them to new tasks by updating all or a subset of parameters. This approach is highly efficient, reduces the need for large labeled datasets, and is particularly relevant for adapting foundation models in NLP, vision, and edge computing. The field is rapidly evolving, with a shift toward methods that balance adaptation quality, computational efficiency, and deployment constraints [8] [23] [24].

2. Latest Trends in Fine-Tuning Neural Networks

Parameter-Efficient Fine-Tuning

Recent research overwhelmingly favors parameter-efficient fine-tuning (PEFT) methods, which update only a small subset of model parameters—often adapters or low-rank matrices—while keeping the majority of the pre-trained model frozen. Key methods include:

- **LoRA (Low-Rank Adaptation):** Injects trainable low-rank matrices into transformer layers, achieving near full fine-tuning performance with drastically fewer parameters and reduced computational cost [1] [2] [3].
- **QLoRA:** Extends LoRA with quantization, enabling fine-tuning of very large models (e.g., 70B parameters) on single GPUs without sacrificing accuracy [25] [26].
- **Adapter-Based Methods:** Insert small trainable modules (adapters) into each layer, allowing efficient domain adaptation and easy parameter sharing across tasks [1] [2].

Impact: These methods are widely adopted in NLP, vision, and remote sensing, enabling efficient deployment on edge devices and in resource-constrained environments [3] [23] [24].

Prompt-Based and Adapter-Based Fine-Tuning

- Prompt Tuning: Optimizes a small set of prompt tokens prepended to the input, guiding the model toward the target task. Particularly effective in few-shot and zero-shot settings, but less robust in complex or unstructured domains [4] [5].
- Comparative Performance: Adapter-based methods (e.g., LoRA, QLoRA) generally outperform prompt tuning in domain-specific and complex tasks, offering better parameter efficiency and stability [2] [5] [27].
- Hybrid Approaches: Combining adapters and prompt tuning leverages the strengths of both, optimizing for both flexibility and domain alignment [2] [6].

Reinforcement Learning-Based Fine-Tuning

- RLHF (Reinforcement Learning from Human Feedback): Uses human preferences to train a reward model, which then guides further fine-tuning, improving model alignment and explainability [9] [11].
- DPO (Direct Preference Optimization): A more stable and computationally efficient alternative to RLHF, focusing on direct optimization of preference-based objectives [10].
- Sample Efficiency: RL-based methods require fewer labeled samples than supervised fine-tuning, making them attractive for aligning large models with human values [9] [28].

Hardware-Aware Fine-Tuning

- Algorithm-Hardware Co-Design: Integrates hardware constraints (e.g., RRAM/SRAM endurance, analog drift) into the fine-tuning process, optimizing for energy efficiency, latency, and deployment on edge devices [12] [13] [14].
- Quantization & In-Memory Computing: Techniques like 8-bit quantization (FP8, Posit8) and analog in-memory computing enable efficient fine-tuning and inference with minimal accuracy loss [13] [29].
- Drift Mitigation: Hardware-aware training and drift compensation techniques reduce the impact of analog memory drift, ensuring stable long-term deployment [12] [29].

Multi-Source and Ensemble Fine-Tuning

- Multi-Source Fine-Tuning: Leverages multiple datasets or modalities, improving robustness and generalization, especially in biomedical signal processing [15] [16].
- Ensemble Strategies: Combine outputs from multiple fine-tuned models (e.g., hard/soft voting), further enhancing accuracy and robustness [16] [30].

Synthesis:

The field is converging on parameter-efficient, hardware-aware, and hybrid fine-tuning methods, with strong evidence supporting their efficiency, scalability, and adaptability across domains. RL-based and privacy-preserving approaches are also gaining traction, addressing alignment and confidentiality challenges.

3. Classification of Fine-Tuning Algorithms

Algorithm Types and Application Domains

Algorithm Type	Description & Trade-offs	Application Domains	References
Supervised Fine-Tuning	Updates all or most parameters using labeled data; high accuracy, high resource cost	NLP, Vision, Biomedical	[31] [32] [33]
Parameter-Efficient (PEFT)	Updates small modules (adapters, LoRA, BitFit); efficient, less overfitting	NLP, Vision, Edge AI	[31] [32] [33]
Prompt/Instruction Tuning	Optimizes prompts or instructions; lightweight, excels in few-shot/zero-shot	NLP, Medical Imaging	[4]
Quantization-Aware	Fine-tunes quantized models for efficient deployment	Edge AI, In-memory Computing	[13] [34]
RL-Based	Uses reward models/human feedback for alignment	LLMs, Medical, Legal	[9] [10]
Multi-Source/Ensemble	Integrates multiple datasets/models for robustness	Biomedical, Multimodal Tasks	[15] [16]
Privacy-Preserving	Differential privacy, federated learning, BitLoRA	Healthcare, Finance, On-device AI	[17] [18] [19]

Single-Source vs. Multi-Source Fine-Tuning

- **Single-Source:** Simpler, less resource-intensive, but limited generalization and transferability [35] [36].
- **Multi-Source:** Integrates diverse data/modalities, improves accuracy and robustness, but increases model complexity and computational cost [37] [38].

- Trade-offs: Multi-source approaches are preferable for cross-subject and cross-domain tasks, especially in biomedical signal processing, but require advanced fusion and domain adaptation strategies [38] [39].

Optimization and Technical Strategies

- Learning Rate Scheduling: Smaller rates for pre-trained layers, larger for new layers; non-uniform schedules reduce catastrophic forgetting [22] [40].
- Parameter Initialization: Pre-trained weights for convolutional layers, random for others; facilitates faster convergence [41] [42].
- Layer Freezing: Early layers often frozen to retain general features; later layers fine-tuned for task adaptation [43] [44].
- Hyperparameter Tuning: Critical for optimal performance; includes learning rate, batch size, optimizer, epochs [45].

Fusion and Ensemble Strategies

- Advanced Fusion: Multi-kernel embedding, attention-based, hierarchical, and cross-modal fusion address modality heterogeneity in multi-source fine-tuning [37] [46].
- Ensemble Learning: Bagging, boosting, stacking, and voting strategies improve robustness and transferability, especially in biomedical and multimodal tasks [16] [30].

Synthesis:

Fine-tuning algorithms are increasingly specialized by technical approach and application domain. Parameter-efficient, multi-source, and ensemble strategies are favored for their adaptability and robustness, while optimization techniques (learning rate, initialization, freezing) are critical for maximizing performance and mitigating forgetting.

4. Advancements and Emerging Techniques in Fine-Tuning

Adapter-Based Fine-Tuning Innovations

- LoRA/QLoRA: Continued improvements in quantization, rank adaptation, and memory efficiency (e.g., RoLoRA, SBoRA, TeamLoRA, PiSSA, NOLA) enable fine-tuning of massive models with minimal resource usage and high accuracy [1] [47] [48] [49] [50].
- Quantization-Aware Variants: QA-LoRA, Q-BLoRA, and others enable efficient INT4/INT8 fine-tuning, supporting direct deployment on low-precision hardware [51] [52].
- NormAL LoRA: Dynamically selects optimal ranks, reducing adapter parameters by 37% while preserving performance [53].

Hybrid Fine-Tuning Approaches

- Adapters + Prompt Tuning: Hybrid methods (e.g., SK-Tuning) combine the adaptability of prompts with the stability of adapters, excelling in few-shot, zero-shot, and domain-specific tasks [6] [7].

- Biomedical/Clinical NLP: Hybrid approaches are particularly effective in specialized domains, balancing efficiency and domain alignment [54] [55].

Hardware-Aware Accelerator Designs

- Analog Memory Drift Mitigation: Hardware-aware fine-tuning and drift compensation techniques reduce accuracy loss in analog in-memory computing (AIMC) systems [12] [29].
- Quantization Formats: 8-bit FP8/Posit8 and other formats reduce area and power consumption with minimal accuracy loss [13].
- Algorithm-Hardware Co-Design: Frameworks like CoXplorer and ANNA integrate model compression and hardware design, optimizing throughput and energy efficiency for edge deployment [14] [56].

Multi-Task Learning and Domain Adaptation

- Parameter-Efficient Multi-Task Fine-Tuning: Shared hypernetworks and TeamLoRA enable efficient knowledge sharing and adaptation across tasks [49] [57].
- Task-Agnostic Domain Adaptation: Methods like TADA retrain embeddings and tokenizers, improving efficiency and generalization without additional parameters [58].
- Ethical Considerations: Interpretability frameworks and fairness-aware fine-tuning are increasingly integrated, especially in sensitive domains [59] [60].

Approximate Computing Paradigms

- Software-Level: Neural architecture search, hyperparameter tuning, and partial fine-tuning optimize accuracy-efficiency trade-offs [61] [62].
- Hardware-Level: Approximate arithmetic units (adders, multipliers) and in-memory computing reduce power and latency, with neural networks tolerating minor inaccuracies [63] [64].
- Co-Design: Joint optimization of approximate hardware and network architecture maximizes efficiency with minimal accuracy loss [65].

Synthesis:

Recent advancements focus on maximizing efficiency and adaptability through innovations in adapter-based methods, hybrid strategies, hardware-aware design, and approximate computing. These techniques are critical for deploying large models in resource-constrained and privacy-sensitive environments.

5. Research Gaps and Underexplored Areas in Fine-Tuning

Mitigating Catastrophic Forgetting

- IMSM: Interweaves memories from pre-trained and fine-tuned parameters, regulating their contributions to reduce forgetting [21].

- Half Fine-Tuning (HFT): Freezes half the parameters, reducing training time by ~30% and improving knowledge retention [20].
- Bayesian PEFT: Uses Laplace approximations to regularize parameter shifts, outperforming diagonal methods in stability-plasticity trade-off [66].
- Learning Rate Scheduling: Non-uniform schedules and parameter averaging reduce forgetting more effectively than flat rates [22] [67].

Privacy-Preserving Fine-Tuning

- Differential Privacy + PEFT: Combines noise injection with parameter-efficient methods (e.g., DP-BiTfIT, DP-FedLoRA) for confidential data adaptation [17] [19] [68].
- Federated Learning: Enables local updates without sharing raw data, critical for healthcare and finance [18] [69].
- BitLoRA: Integrates quantization and adapter tuning for privacy-preserving, memory-efficient on-device adaptation [70].

Parameter-Efficient Frameworks for Forgetting Mitigation

- IMSM, Bayesian PEFT, LiLoRA, Half Fine-Tuning: All demonstrate strong performance in mitigating catastrophic forgetting in large language models, with minimal resource overhead [20] [21] [71].

Learning Rate Scheduling and Knowledge Retention

- Non-uniform Schedules: Outperform flat rates in reducing forgetting [22].
- Parameter Averaging: Reduces forgetting to as low as 3.4%, outperforming LoRA and other methods [67].

Future Directions and Open Challenges

- Hardware Constraints: Need for robust, unified algorithm-hardware co-design frameworks for nonvolatile and analog memory systems [8] [72].
- Model Compression: Balancing compression with accuracy for real-time, energy-efficient deployment remains underexplored [72] [73].
- Interpretability & Privacy: Black-box nature and privacy concerns are often overlooked; more research needed on explainable and privacy-preserving fine-tuning [74].
- Domain Adaptation: Automated, efficient domain adaptation and negative transfer mitigation in multi-source fine-tuning [75] [76].

Synthesis:

Despite significant progress, challenges remain in catastrophic forgetting, privacy, hardware constraints, and interpretability. Addressing these gaps is essential for robust, scalable, and trustworthy fine-tuning in real-world applications.

6. Real-World Implications

- **Edge Deployment:** Parameter-efficient and hardware-aware fine-tuning enable large models to run on edge devices, supporting real-time applications in healthcare, IoT, and autonomous systems [12] [13].
- **Domain Adaptation:** Fine-tuning allows rapid adaptation of foundation models to specialized domains (e.g., legal, medical, code), improving accuracy and reducing hallucinations [77] [78].
- **Privacy & Security:** Differential privacy and federated learning ensure confidential data can be used for model adaptation without compromising user privacy [17] [18].
- **Continual Learning:** Catastrophic forgetting mitigation is critical for models that must adapt over time without losing prior knowledge, especially in dynamic environments [20] [21].
- **Resource Efficiency:** Approximate computing and quantization reduce energy and memory requirements, making AI accessible on low-power devices [13] [63].

7. Future Research Directions

- **Unified Algorithm-Hardware Co-Design:** Develop frameworks that jointly optimize fine-tuning algorithms and hardware architectures for emerging memory technologies.
- **Continual and Lifelong Learning:** Robust, scalable solutions for catastrophic forgetting, especially in cross-domain and low-resource settings.
- **Privacy-Preserving Adaptation:** Integrate differential privacy, federated learning, and parameter-efficient methods for secure, efficient fine-tuning.
- **Automated Domain Adaptation:** Efficient, automated strategies for negative transfer mitigation and domain adaptation in multi-source fine-tuning.
- **Interpretability and Trust:** Enhance transparency and explainability in fine-tuned models, especially for high-stakes applications.

Conclusion:

Fine-tuning neural networks is rapidly evolving, with parameter-efficient, hardware-aware, and privacy-preserving methods at the forefront. These advances are enabling the practical deployment of large models across diverse domains and resource-constrained environments. Addressing current research gaps in catastrophic forgetting, privacy, hardware integration, and interpretability will be crucial for the next generation of adaptive, trustworthy AI systems.

References

1. Koohpayegani, S.A., Navaneet, K.L., Nooralinejad, P., (...), Pirsavash, H. (2024) NOLA: COMPRESSING LORA USING LINEAR COMBINATION OF RANDOM BASIS. *12th International Conference on Learning Representations, ICLR 2024*. <https://www.scopus.com/pages/publications/85198191425>
2. Al-Dulaimi, A., Akel, A., Ghwanmeh, S., (...), Ibrahim, A. (2025) Efficient Fine-Tuning Strategies for Domainspecific Large Language Models. *3rd International Conference on Business Analytics for Technology and Security, ICBATS 2025*. <https://doi.org/10.1109/ICBATS66542.2025.11258493>
3. Singh, K., Deshpande, S., Patwardhan, S. (2025) Refining the Giants: A Comprehensive Review of Fine-Tuning Strategies for Large Language Models. *Lecture Notes in Networks and Systems*. https://doi.org/10.1007/978-981-97-7839-3_5
4. Zu, W., Xie, S., Zhao, Q., (...), Ma, L. (2024) Embedded prompt tuning: Towards enhanced calibration of pretrained models for medical images. *Medical Image Analysis*. <https://doi.org/10.1016/j.media.2024.103258>
5. Gajulamandyam, D.K., Veerla, S., Emami, Y., (...), Shim, S. (2025) Domain Specific Finetuning of LLMs Using PEFT Techniques. *2025 IEEE 15th Annual Computing and Communication Workshop and Conference, CCWC 2025*. <https://doi.org/10.1109/CCWC62904.2025.10903789>
6. Prottasha, Nusrat Jahan, Mahmud, Asif, Sobuj, Md. Shohanur Islam, (...), Garibay, Ozlem Ozmen (2099) Parameter-efficient fine-tuning of large language models using semantic knowledge tuning. *Scientific Reports*. <https://doi.org/10.1038/s41598-024-75599-4>
7. Shen, Mengfan, Song, Kangqi, Wang, Xindi, (...), Han, Ziqiang (2026) A Domain-Adapted Pipeline for Structured Information Extraction from Police Incident Announcements on Social Media. *Information Processing & Management*. <https://doi.org/10.1016/j.ipm.2026.104788>
8. Mu, C., Huang, Z., Jiang, H., (...), Chen, C. (2024) A 28nm 76.25TOPS/W RRAM/SRAM-Collaborative CIM Fine-Tuning Accelerator with RRAM-Endurance/Latency-Aware Weight Allocation for CNN and Transformer. *2024 IEEE Asian Solid-State Circuits Conference, A-SSCC 2024*. <https://doi.org/10.1109/A-SSCC60305.2024.10848596>
9. Drazen, Jeffrey M. (2024) Medical Artificial Intelligence and Human Values. *The New England Journal of Medicine*. <https://doi.org/10.1056/NEJMr2214183>
10. Avshalumov, M., Volovikova, Z., Yudin, D., Panov, A. (2026) Say it better: RL-based prompt tuning for enhancing open-vocabulary recognition. *Neurocomputing*. <https://doi.org/10.1016/j.neucom.2026.133353>
11. Jing, P., Lee, K., Zhang, Z., (...), Yang, G. (2026) Reason like a radiologist: Chain-of-thought and reinforcement learning for verifiable report generation. *Medical Image Analysis*. <https://doi.org/10.1016/j.media.2025.103910>

12. Chen, A., Ambrogio, S., Narayanan, P., (...), Burr, G.W. (2025) Demonstration of transformer-based ALBERT model on a 14nm analog AI inference chip. *Nature Communications*. <https://doi.org/10.1038/s41467-025-63794-4>
13. Yu, J., Prabhu, K., Urman, Y., (...), Raina, P. (2024) 8-bit Transformer Inference and Fine-tuning for Edge Accelerators. *International Conference on Architectural Support for Programming Languages and Operating Systems - ASPLOS*. <https://doi.org/10.1145/3620666.3651368>
14. Krestinskaya, O., Fouda, M.E., Eltawil, A.M., Salama, K.N. (2026) Joint Hardware-Workload Co-Optimization for In-Memory Computing Accelerators. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2026.3671360>
15. Hoshino, T., Kanoga, S., Tsubaki, M., Aoyama, A. (2024) Comparison of fine-tuned single-source and multi-source approaches to surface electromyogram pattern recognition. *Biomedical Signal Processing and Control*. <https://doi.org/10.1016/j.bspc.2024.106261>
16. Hoshino, T., Kanoga, S., Tsubaki, M., Aoyama, A. (2024) Comparison of fine-tuned single-source and multi-source approaches to surface electromyogram pattern recognition. *Biomedical Signal Processing and Control*. <https://doi.org/10.1016/j.bspc.2024.106261>
17. Sarkar, A.R., Sarmin, F.J., Mouheb, D., (...), Mohammed, N. (2026) Assessment of differentially private fine-tuning of large language models for synthetic clinical note generation. *Intelligent Systems with Applications*. <https://doi.org/10.1016/j.iswa.2026.200659>
18. Ali, S., Talpur, D.B., Humayun, M., (...), Yaqoob, J. (2026) The double-edged sword: LLM operations (LLMOps) security in the cloud- a comprehensive review. *Neurocomputing*. <https://doi.org/10.1016/j.neucom.2026.133313>
19. Bu, Z., Wang, Y.-X., Zha, S., Karypis, G. (2024) Differentially Private Bias-Term Fine-tuning of Foundation Models. *Proceedings of Machine Learning Research*. <https://www.scopus.com/pages/publications/85199628876>
20. Hui, T., Zhang, Z., Wang, S., (...), Wu, H. (2025) HFT: Half Fine-Tuning for Large Language Models. *Proceedings of the Annual Meeting of the Association for Computational Linguistics*. <https://doi.org/10.18653/v1/2025.acl-long.626>
21. Song, X., Xue, Z., He, G., (...), Lu, W. (2025) Interweaving Memories of a Siamese Large Language Model. *Proceedings of the AAAI Conference on Artificial Intelligence*. <https://doi.org/10.1609/aaai.v39i23.34701>
22. Kenneweg, P., Schulz, A., Schröder, S., Hammer, B. (2022) Intelligent Learning Rate Distribution to Reduce Catastrophic Forgetting in Transformers. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*. https://doi.org/10.1007/978-3-031-21753-1_25
23. Yu, B.X.B., Chang, J., Wang, H., (...), Chen, C.W. (2024) Visual Tuning. *ACM Computing Surveys*. <https://doi.org/10.1145/3657632>

24. Feng, J., Shi, Y. (2025) A bibliometric studies of pre-trained model and fine-tune method. *Procedia Computer Science*. <https://doi.org/10.1016/j.procs.2025.08.159>
25. Santos, D., Nogueira, V.B., Quaresma, P. (2025) Parameter Efficient Fine-Tuning of LLMs: Application to Machine Translation from English to Portuguese. *Proceedings - 2025 4th International Conference on Computer Technologies, ICTech 2025*. <https://doi.org/10.1109/ICTech66294.2025.00014>
26. Hanindhito, B., Patel, B., John, L.K. (2025) Large Language Model Fine-tuning with Low-Rank Adaptation: A Performance Exploration. *ICPE 2025 - Proceedings of the 16th ACM/SPEC International Conference on Performance*. <https://doi.org/10.1145/3676151.3719377>
27. Hu, Z., Wang, L., Lan, Y., (...), Lee, R.K.-W. (2023) LLM-Adapters: An Adapter Family for Parameter-Efficient Fine-Tuning of Large Language Models. *EMNLP 2023 - 2023 Conference on Empirical Methods in Natural Language Processing, Proceedings*. <https://doi.org/10.18653/v1/2023.emnlp-main.319>
28. Viglia, G., Adler, S.J., Lancelot-Miltgen, C.L., Sarstedt, M. (2024) The use of synthetic data in tourism. *Annals of Tourism Research*. <https://doi.org/10.1016/j.annals.2024.103819>
29. Wu, Z., Su, X., Rasch, M.J., Li, N. (2026) Methods for Setting Device Specifications for Analog In-Memory Computing Inference. *Advanced Intelligent Systems*. <https://doi.org/10.1002/aisy.202501355>
30. Suzaimi, N.D., Abuhabib, A., Darwish, M., (...), Hamzah, S. (2026) Machine learning in industrial wastewater treatment: Critical insights into modelling approaches and digital futures. *Journal of Industrial and Engineering Chemistry*. <https://doi.org/10.1016/j.jiec.2025.12.035>
31. Sai, C.V., Elmitwally, N., Rice, I., (...), Schmoor, X. (2025) Fine-Tuning Large Language Models for Domain-Specific Tasks: A Comparative Study on Supervised and Parameter-Efficient Fine-Tuning Techniques for Cybersecurity and IT Support. *Lecture Notes on Data Engineering and Communications Technologies*. https://doi.org/10.1007/978-3-031-91351-8_14
32. Ramesh, G., Shrikanth, N.G., Mahadev, K.G., Mohammed, R.N. (2025) Fine-Tuning Techniques for Large Language Models: A Comprehensive survey. *2025 IEEE International Conference on Electronics, Computing and Communication Technologies, CONECCT 2025*. <https://doi.org/10.1109/CONECCT65861.2025.11306881>
33. Zhang, Q., Wang, Y., Wang, H., (...), Chen, H. (2024) Comprehensive Review of Large Language Model Fine-Tuning. *Computer Engineering and Applications*. <https://doi.org/10.3778/j.issn.1002-8331.2312-0035>
34. Liang, Y., Lin, H., Qiu, S., Zhang, Y. (2024) AAT: ADAPTING AUDIO TRANSFORMER FOR VARIOUS ACOUSTICS RECOGNITION TASKS. *ICASSP, IEEE International Conference on Acoustics, Speech and Signal Processing - Proceedings*. <https://doi.org/10.1109/ICASSP48485.2024.10447544>
35. Wei, Y., Cao, L., Dong, Y., Liu, T. (2025) MCNN-CMCA: A multiscale convolutional neural networks with cross-modal channel attention for physiological signal-based mental state recognition. *Digital Signal Processing: A Review Journal*. <https://doi.org/10.1016/j.dsp.2024.104856>

36. Pan, J., He, Z., Li, Z., (...), Qiu, L. (2020) A review of multimodal emotion recognition. *CAA/ Transactions on Intelligent Systems*. <https://doi.org/10.11992/tis.202001032>
37. Ma, F., Luo, H., Guo, Z., (...), Zhang, P. (2025) Multilevel Spatiotemporal Modeling Network for Emotion Recognition With Multimodal Physiological Signals. *IEEE Transactions on Instrumentation and Measurement*. <https://doi.org/10.1109/TIM.2025.3618736>
38. Yuan, K., Hou, L., Huang, Y. (2023) Multi-source domain adaptive dictionary learning and sparse representation approach for electroencephalogram-based emotion recognition. *Journal of Nanjing University of Information Science and Technology*. <https://doi.org/10.13878/j.cnki.jnuist.20220623001>
39. Chen, C., Chen, L., Zhang, X. (2023) Mental Workload Recognition Based on Multi-Source Deep Domain Adaptation. *Huadong Ligong Daxue Xuebao/Journal of East China University of Science and Technology*. <https://doi.org/10.14135/j.cnki.1006-3080.20220723002>
40. Kounalakis, T., Triantafyllidis, G.A., Nalpantidis, L. (2019) Deep learning-based visual recognition of rumex for robotic precision farming. *Computers and Electronics in Agriculture*. <https://doi.org/10.1016/j.compag.2019.104973>
41. Jiang, M.-X., Deng, C., Shan, J.-S., (...), Sun, X. (2019) Hierarchical multi-modal fusion FCN with attention model for RGB-D tracking. *Information Fusion*. <https://doi.org/10.1016/j.inffus.2018.09.014>
42. Yang, T., Long, X., Arun Kumar, A.K., (...), Tong, C. (2018) Deep detection network for real-life traffic sign in vehicular networks. *Computer Networks*. <https://doi.org/10.1016/j.comnet.2018.02.026>
43. Wang, L., Yi, S., Yu, Y., (...), Samali, B. (2024) Automated ultrasonic-based diagnosis of concrete compressive damage amidst temperature variations utilizing deep learning. *Mechanical Systems and Signal Processing*. <https://doi.org/10.1016/j.ymssp.2024.111719>
44. Talaei, S., Zhu, X., Li, J., (...), Chan, T.H.T. (2023) Transfer learning based bridge damage detection: Leveraging time-frequency features. *Structures*. <https://doi.org/10.1016/j.istruc.2023.105052>
45. Han, D., Liu, Q., Fan, W. (2018) A new image classification method using CNN transfer learning and web data augmentation. *Expert Systems with Applications*. <https://doi.org/10.1016/j.eswa.2017.11.028>
46. He, X., Huang, J., Fu, Z., (...), Wu, D. (2025) A Multi-Kernel Embedding Fusion Framework for Physiological Signal Based Emotion Recognition. *IEEE Transactions on Affective Computing*. <https://doi.org/10.1109/TAFFC.2025.3562905>
47. Huang, X., Liu, Z., Liu, S.-Y., Cheng, K.-T. (2024) RoLoRA: Fine-tuning Rotated Outlier-free LLMs for Effective Weight-Activation Quantization. *EMNLP 2024 - 2024 Conference on Empirical Methods in Natural Language Processing, Findings of EMNLP 2024*. <https://doi.org/10.18653/v1/2024.findings-emnlp.444>
48. Po, L.-M., Liu, Y., Wu, H., (...), Li, K. (2026) SBoRA: Low-Rank Adaptation with Regional Weight Updates. *Communications in Computer and Information Science*. https://doi.org/10.1007/978-981-96-7008-6_28

49. Lin, T., Liu, J., Zhang, W., (...), Zhuang, Y. (2025) TeamLoRA: Boosting Low-Rank Adaptation with Expert Collaboration and Competition. *Proceedings of the Annual Meeting of the Association for Computational Linguistics*. <https://doi.org/10.18653/v1/2025.acl-long.669>
50. Meng, F., Wang, Z., Zhang, M. (2024) PiSSA: Principal Singular Values and Singular Vectors Adaptation of Large Language Models. *Advances in Neural Information Processing Systems*. <https://www.scopus.com/pages/publications/105000765143>
51. Shen, A., Lai, Z., Wang, Q., (...), Li, J. (2025) Accurate and Efficient Fine-Tuning of Quantized Large Language Models Through Optimal Balance in Adaptation. *Transactions of the Association for Computational Linguistics*. <https://doi.org/10.1162/TACL.a.23>
52. Xu, Y., Xie, L., Gu, X., (...), Tian, Q. (2024) QA-LORA: QUANTIZATION-AWARE LOW-RANK ADAPTATION OF LARGE LANGUAGE MODELS. *12th International Conference on Learning Representations, ICLR 2024*. <https://www.scopus.com/pages/publications/85192836823>
53. Aastik, Topu, M., Kulkarni, C., Sahu, P.P. (2025) NormAL LoRA: What is the perfect size?. *EMNLP 2025 - 2025 Conference on Empirical Methods in Natural Language Processing, Findings of EMNLP 2025*. <https://doi.org/10.18653/v1/2025.findings-emnlp.1074>
54. Peng, C., Yang, X., Smith, K.E., (...), Wu, Y. (2024) Model tuning or prompt Tuning? a study of large language models for clinical concept and relation extraction. *Journal of Biomedical Informatics*. <https://doi.org/10.1016/j.jbi.2024.104630>
55. Villena, Fabián, Bravo-Marquez, Felipe, Dunstan, Jocelyn (2025) NLP modeling recommendations for restricted data availability in clinical settings. *BMC Medical Informatics and Decision Making*. <https://doi.org/10.1186/s12911-025-02948-2>
56. Li, C., Zhang, K., Li, Y., (...), Qian, L. (2023) ANNA: Accelerating Neural Network Accelerator through software-hardware co-design for vertical applications in edge systems. *Future Generation Computer Systems*. <https://doi.org/10.1016/j.future.2022.10.001>
57. Mahabadi, R.K., Ruder, S., Dehghani, M., Henderson, J. (2021) Parameter-efficient multi-task fine-tuning for transformers via shared hypernetworks. *ACL-IJCNLP 2021 - 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing, Proceedings of the Conference*. <https://doi.org/10.18653/v1/2021.acl-long.47>
58. Hung, C.-C., Lange, L., Strötgen, J. (2023) TADA: Efficient Task-Agnostic Domain Adaptation for Transformers. *Proceedings of the Annual Meeting of the Association for Computational Linguistics*. <https://doi.org/10.18653/v1/2023.findings-acl.31>
59. Matan, P., Velvizhy, P. (2025) A Comprehensive Review of Supervised Fine-Tuning for Large Language Models in Creative Applications and Content Moderation. *Proceedings of 8th International Conference on Inventive Computation Technologies, ICICT 2025*. <https://doi.org/10.1109/ICICT64420.2025.11005111>
60. Sitaraman, H., Jayavel, A., Gnanasekaran, S., (...), Madsen, D.Ø. (2025) Interpretable and ethical learning assessment transformer (IELAT): an explainable transformer model for personalized student assessments. *Cogent Education*. <https://doi.org/10.1080/2331186X.2025.2485508>

61. Tavoni, G., Balasubramanian, V., Gold, J.I. (2019) What is optimal in optimal inference?. *Current Opinion in Behavioral Sciences*. <https://doi.org/10.1016/j.cobeha.2019.07.008>
62. Aras, S., Aras, E., Gedikli, E., Kahraman, H.T. (2025) Multi-objective CNN optimization: A robust framework for automated model design. *Information Sciences*. <https://doi.org/10.1016/j.ins.2025.122468>
63. Askarizadeh, M.J., Castro-Godínez, J., Farahmand, E., (...), Salazar-García, C. (2026) Robust DCNN: The impact of approximate multipliers in defending against adversarial attacks. *Future Generation Computer Systems*. <https://doi.org/10.1016/j.future.2025.108220>
64. Turaka, R., Ravichand, S.R., Anitha, R., (...), Farhaoui, Y. (2023) A novel approach for design energy efficient inexact reverse carry select adders for IoT applications. *Results in Engineering*. <https://doi.org/10.1016/j.rineng.2023.101127>
65. Wu, Y., Yan, Z., Yin, X., (...), Zhuo, C. (2025) ANAS: Software-hardware co-design of approximate neural network accelerators via neural architecture search. *Integration*. <https://doi.org/10.1016/j.vlsi.2025.102469>
66. Chen, H., Garner, P.N. (2024) Bayesian Parameter-Efficient Fine-Tuning for Overcoming Catastrophic Forgetting. *IEEE/ACM Transactions on Audio Speech and Language Processing*. <https://doi.org/10.1109/TASLP.2024.3463395>
67. Plantinga, P., Yoo, J., Girma, A., Dhir, C. (2024) Parameter Averaging Is All You Need To Prevent Forgetting. *Proceedings of 2024 IEEE Spoken Language Technology Workshop, SLT 2024*. <https://doi.org/10.1109/SLT61566.2024.10832275>
68. Xu, H., Shrestha, S., Chen, W., (...), Cai, Z. (2025) DP-FedLoRA: Privacy-Enhanced Federated Fine-Tuning for On-Device Large Language Models. *Proceedings - IEEE International Conference on Data Mining, ICDM*. <https://doi.org/10.1109/ICDM65498.2025.00089>
69. Huang, Zhaojing, Yu, Leping, Fernando Herbozo Contreras, Luis, (...), Kavehei, Omid (2025) *Machine Learning: Science and Technology*. <https://doi.org/10.1088/2632-2153/adaca3>
70. Song, I., Lee, K. (2026) BitLoRA: Quantization-Compatible Adapter Tuning for 1.58-bit LLM in Federated On-Device AI-Agent. *Expert Systems with Applications*. <https://doi.org/10.1016/j.eswa.2026.131397>
71. Che, C., Wang, Z., Yang, P., (...), Shi, Z. (2026) LoRA in LoRA: Towards Parameter-Efficient Architecture Expansion for Continual Visual Instruction Tuning. *Proceedings of the AAAI Conference on Artificial Intelligence*. <https://doi.org/10.1609/aaai.v40i24.39082>
72. Khan, S.A., Shulepina, S., Shulepin, D., Lukmanov, R.A. (2024) Review of algorithmic solutions for deployment of neural networks on lite devices. *Computer Research and Modeling*. <https://doi.org/10.20537/2076-7633-2024-16-7-1601-1619>
73. Cai, M., Su, Y., Wang, B., Zhang, T. (2023) Research on compression pruning methods based on deep learning. *Journal of Physics: Conference Series*. <https://doi.org/10.1088/1742-6596/2580/1/012060>

74. Abdulkadar, J.I., Rama Mohan, C.R., Vijaya Bharati, P.V., (...), Mudholkar, P. (2024) Neural Networks and Deep Learning: Innovations and Applications in Soft Computing. *Proceedings of International Conference on Contemporary Computing and Informatics, IC3I 2024*. <https://doi.org/10.1109/IC3I61595.2024.10829173>
75. Wang, R., Zhang, R., Henao, R. (2022) Wasserstein Uncertainty Estimation for Adversarial Domain Matching. *Frontiers in Big Data*. <https://doi.org/10.3389/fdata.2022.878716>
76. Sun, Y., Chong, N., Ochiai, H. (2022) Feature Distribution Matching for Federated Domain Generalization. *Proceedings of Machine Learning Research*. <https://www.scopus.com/pages/publications/85162219041>
77. Pai, K.-C., Hsu, W.-C. (2026) Fine-tuning small language models for industrial retrieval-augmented generation: Efficiency, factuality, and hallucination analysis. *Computer Standards and Interfaces*. <https://doi.org/10.1016/j.csi.2026.104163>
78. Ramesh, R., Akarsh Thejasvi Raju, M., Reddy, H.V., Nadimpalli, N. (2024) Fine-Tuning Large Language Models for Task Specific Data. *Proceedings - ICNEWS 2024: 2nd International Conference on Networking, Embedded and Wireless Systems: Wireless Technology - Building a Digital World*. <https://doi.org/10.1109/ICNEWS60873.2024.10730913>

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